Problem-Driven Artificial Intelligence for Ghana: Addressing National Computational Bottlenecks through Indigenous Language Classifiers and Edge Metaheuristics

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ABSTRACT

Developing nations face technological exclusion if AI models remain calibrated exclusively to Western linguistic corpora and foreign GPU clouds. This research establishes an evidence-based roadmap identifying the top five computational obstacles confronting Ghana and delivers tangible field-deployable solutions. We evaluated national datasets from the Ghana Statistical Service and the Ministry of Communications and Digitalisation, scoring bottlenecks across impact, urgency, empirical evidence, and technical tractability. This research systematically addresses five national bottlenecks: indigenous language training data scarcity, cloud GPU foreign exchange costs, localized computer science curricula deficits, fragmented government data governance, and agricultural model adoption inertia. As a tangible open-source artifact, we built and validated a deterministic Ghanaian language identification and text classification prototype supporting Akan/Twi, Ewe, Ga, and Ghanaian English in /prototypes/ai-computing/.

Keywords: Artificial Intelligence, Indigenous Languages, Edge Metaheuristics, Data Scarcity, Compute Sovereignty, Ghana

PROBLEM-TO-SOLUTION SUMMARY & OPEN ARTIFACT

This publication presents an empirical pathway connecting established computational and engineering theory to physical, field-validated artifacts addressing five ranked national problems in Ghana. A runnable, tested open-source prototype has been authored and published under an MIT license in the repository directory /prototypes/ai-computing/ with automated unit test suites.

Canonical Web Reader: https://bernardfiagbenu.vercel.app/research/ai-computing